ADVANCED ROBOTIC PERCEPTION / IMAGE PROCESSING AND COMPUTER VISION

HUMAN POSE ESTIMATION

RIKKE GADE, 2026

AALBORG UNIVERSITY
DENMARK

Pose estimation



https://arxiv.org/pdf/1902.09212.pdf

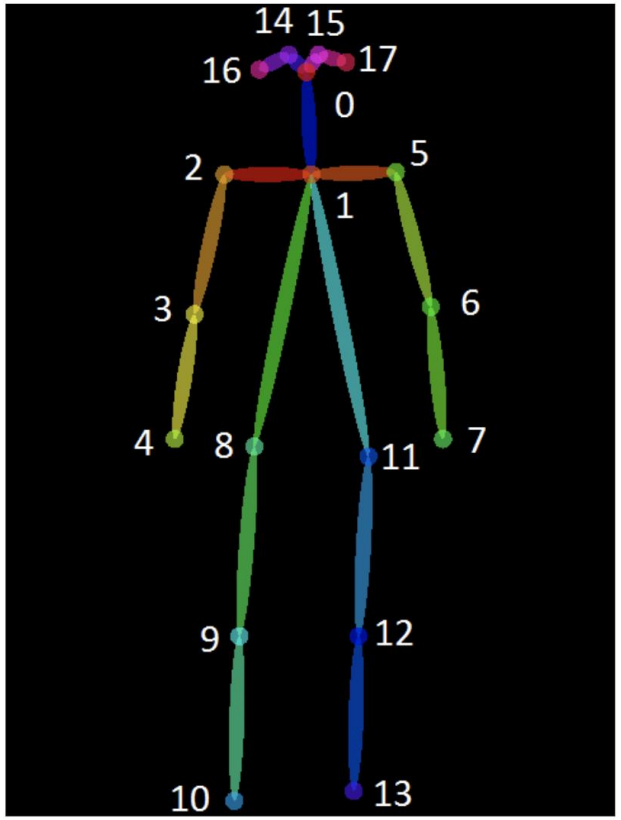
Pose estimation

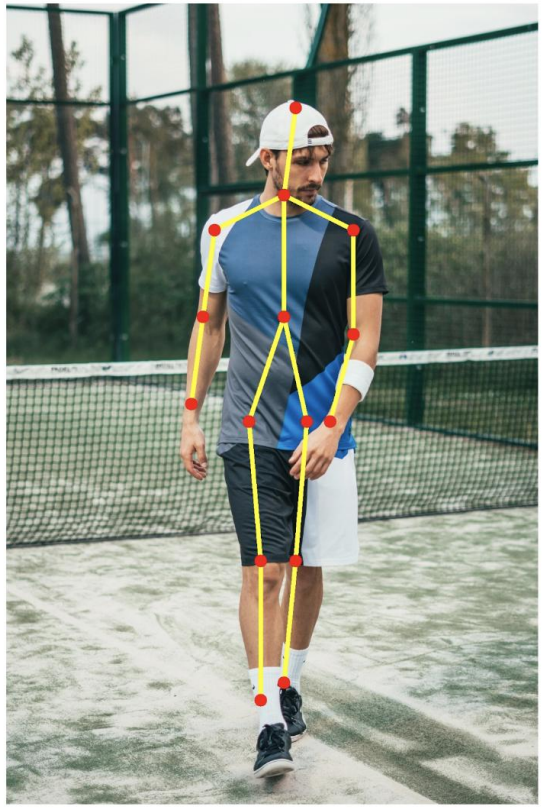
Human pose estimation

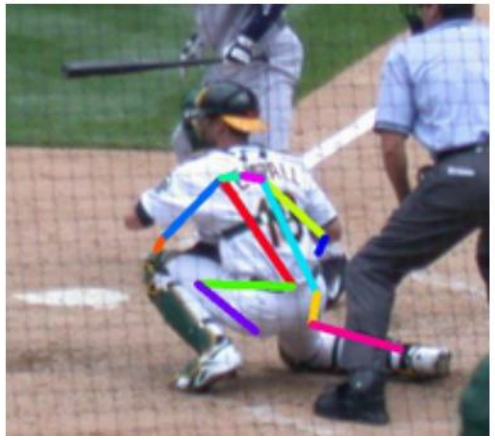
Goal: Estimate the position of each joint in 2D (x,y) / 3D (x,y,z)

Difficulties:

- Many degrees of freedom
- Appearance variations (clothes, body shape)
- (Self-) Occlusions







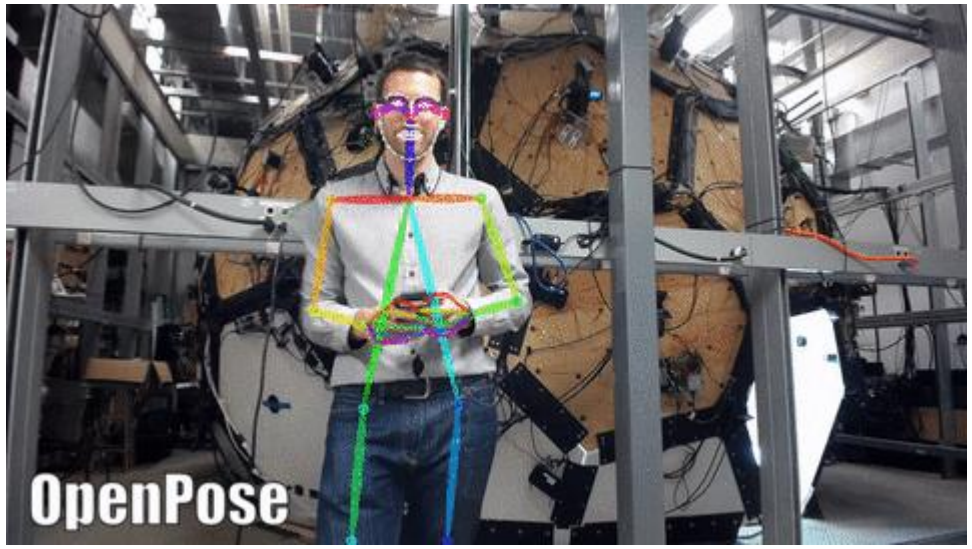


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Pose estimation

Why do want this?

- Robots interacting with humans
- Action recognition (video understanding)
- Gestures
- Gaming
- Rehabilitation
- Performance analysis (sports)
- …



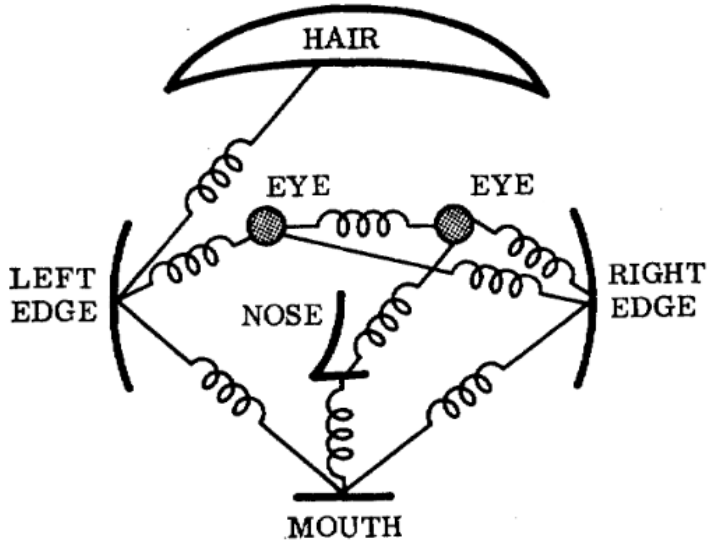
https://github.com/CMU-Perceptual-Computing-Lab/openpose

Pose estimation – 2D

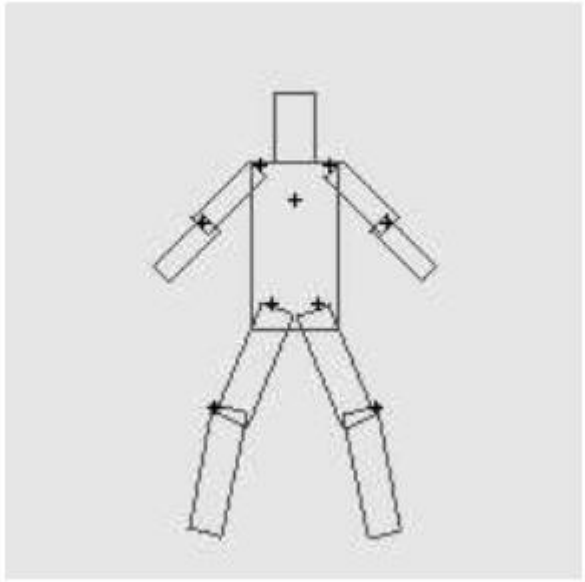
Classic approaches (pre-deep learning):

Pictorial structures:

- Local part templates
- Geometric constraints (springs)



https://ieeexplore.ieee.org/abstract/document/1672195



https://link.springer.com/article/10.1023/B:VISI.0000042934.15159.49

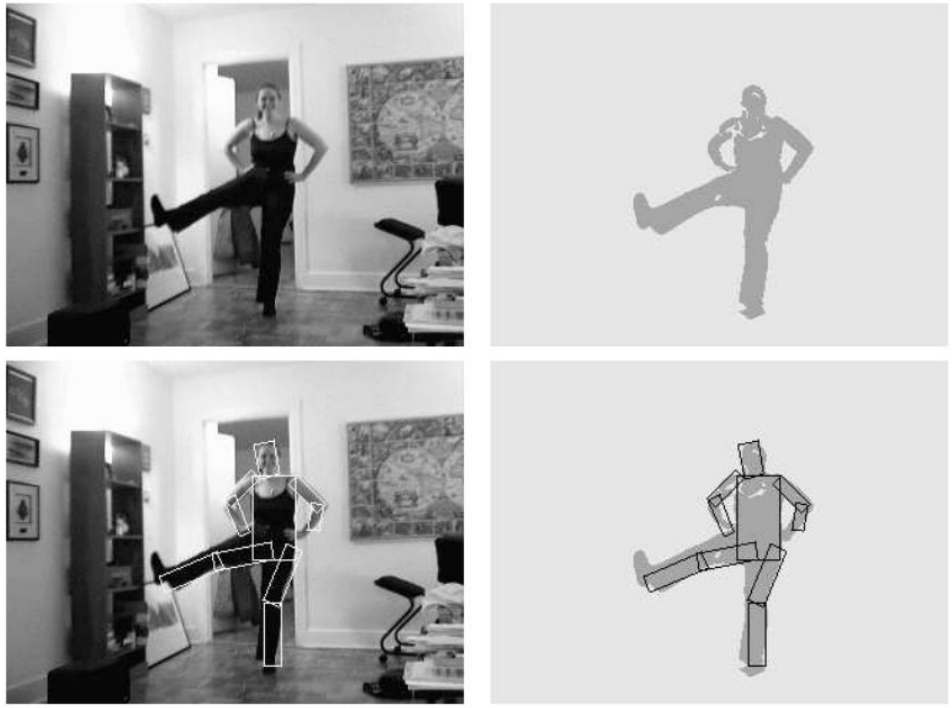
P. Felzenszwalb and D. Huttenlocher, “Pictorial structures for object recognition,”International Journal of Computer Vision, 2005.

M. Fischler and R. Elschlager, “The representation and matching of pictorial structures,”IEEE Transactions on Computers, 1973

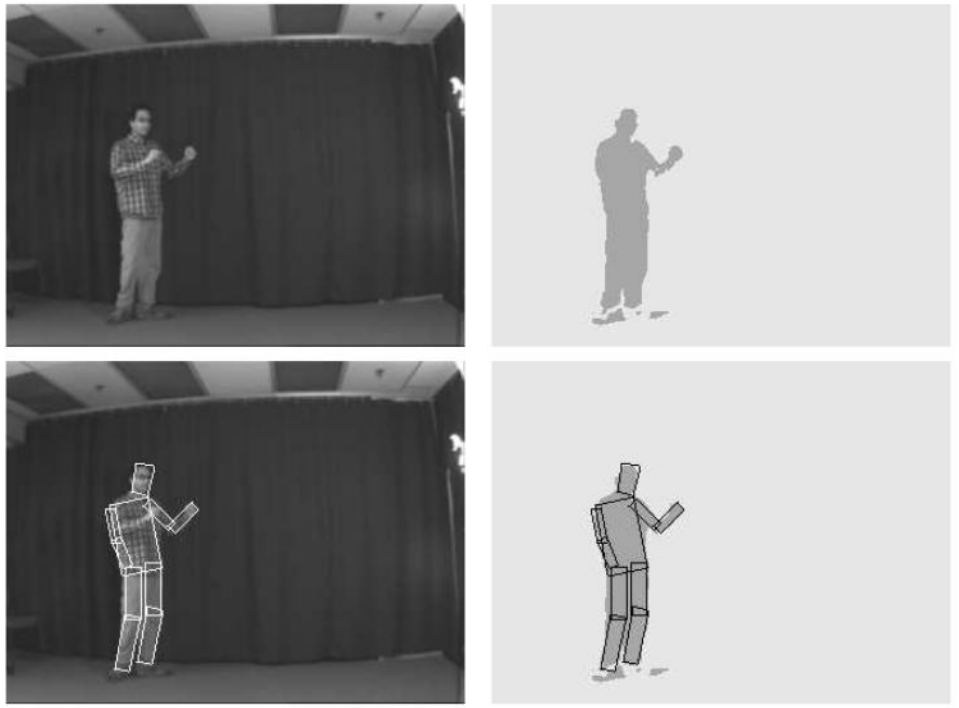
Pose estimation – 2D

Pictorial structures

Good case



Bad case

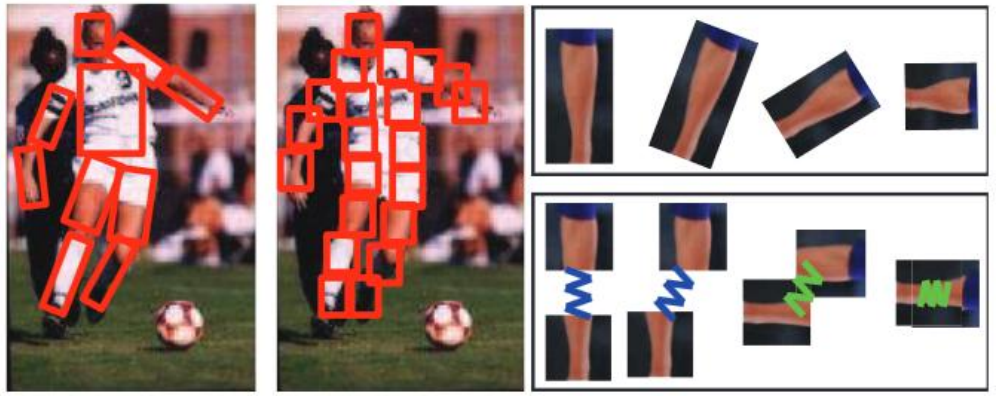


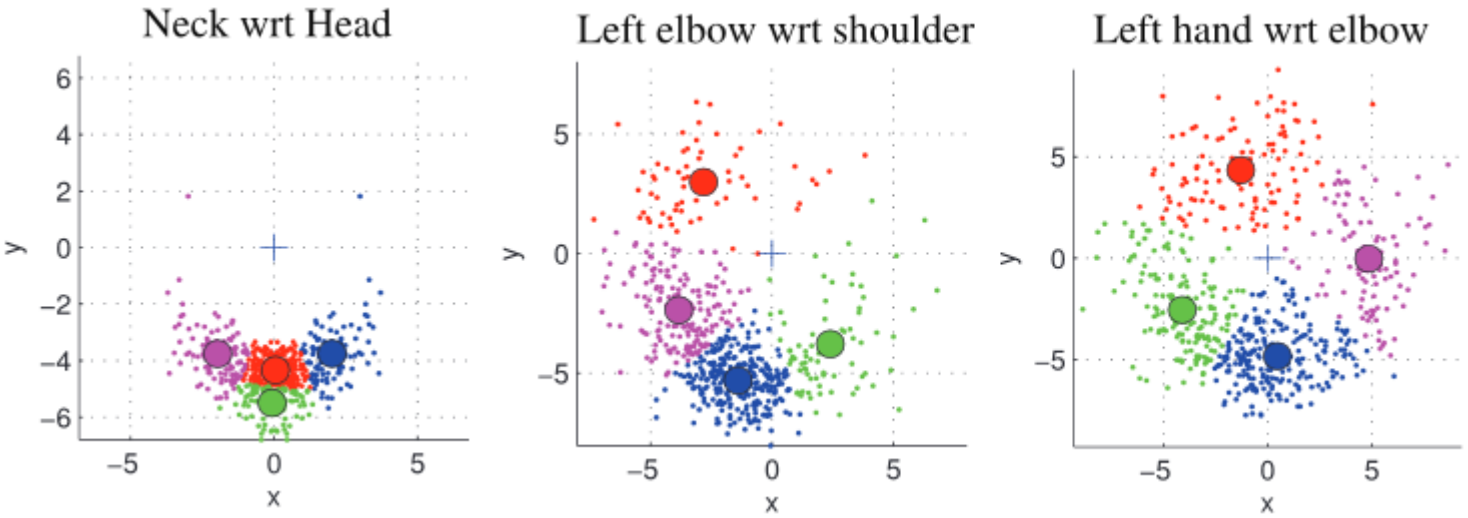
Pose estimation – 2D

Classic approaches (pre-deep learning):

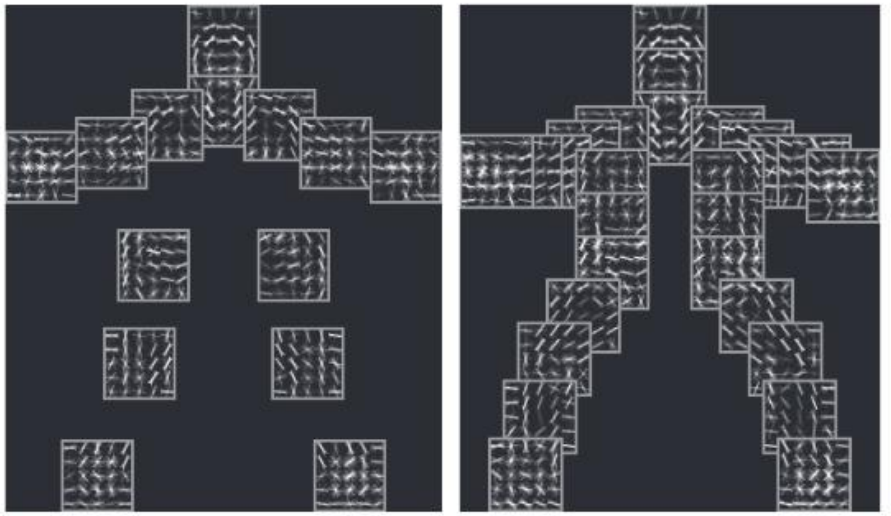
Deformable part models:

- Smaller parts
- Global templates + part templates
- Learned relations



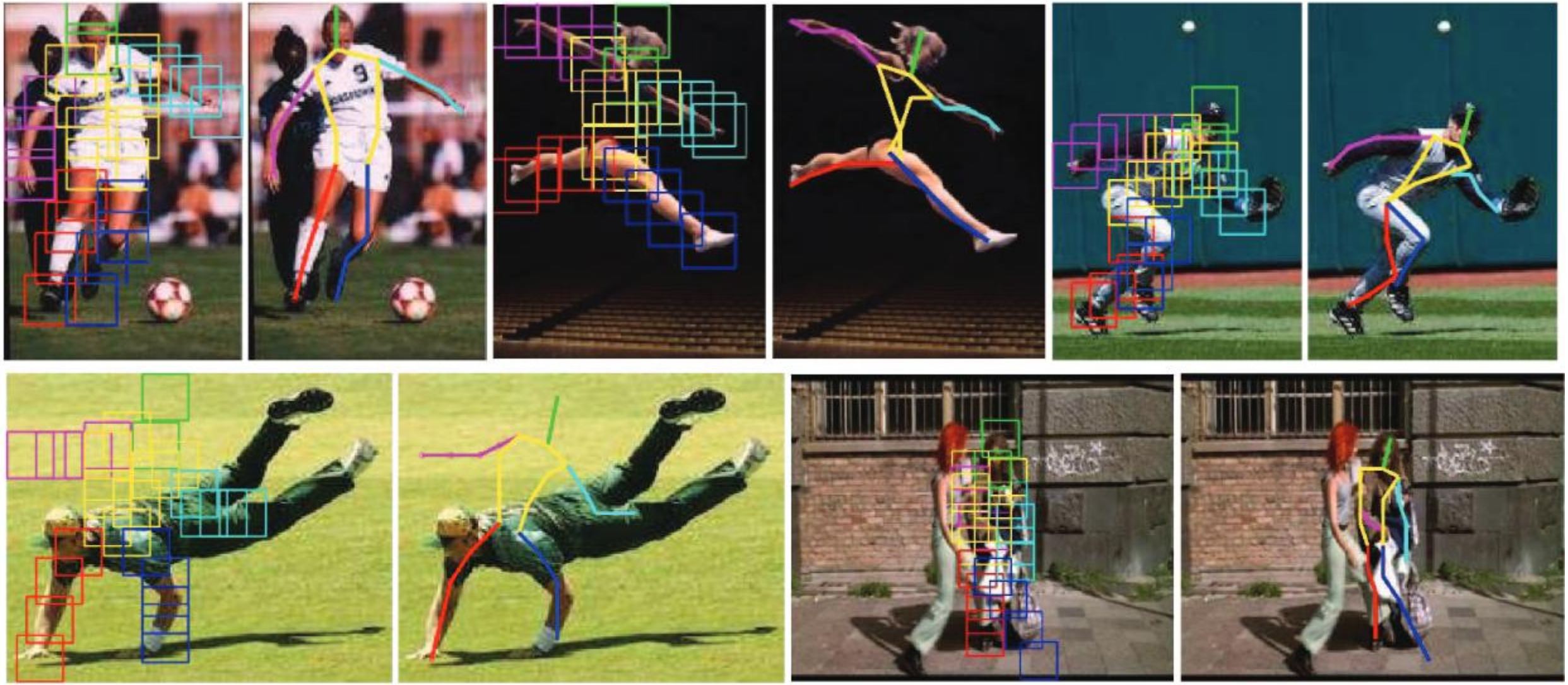


Y. Yang and D. Ramanan, "Articulated Human Detection with Flexible Mixtures of Parts", PAMI, 2013



https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=6380498

Pose estimation – 2D



Pose estimation – 2D

Modern approach: Use deep learning!

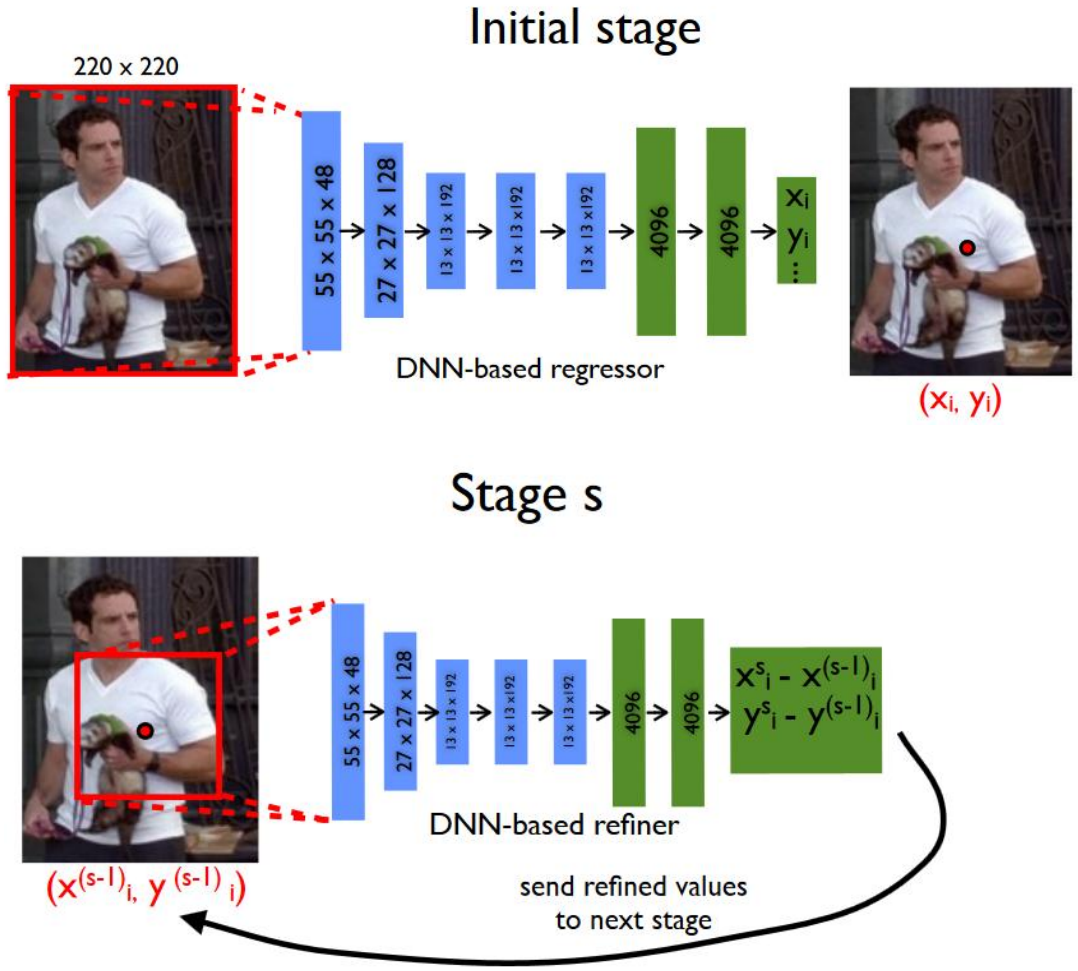
- Train a CNN to locate joint positions



Pose estimation – 2D

DeepPose: Human Pose Estimation via Deep Neural Networks

- First real DNN-based human pose estimation method
- Removed the need for hand-crafted features and model topology
- CNN-based regression problem



https://arxiv.org/pdf/1312.4659.pdf

Pose estimation – 2D

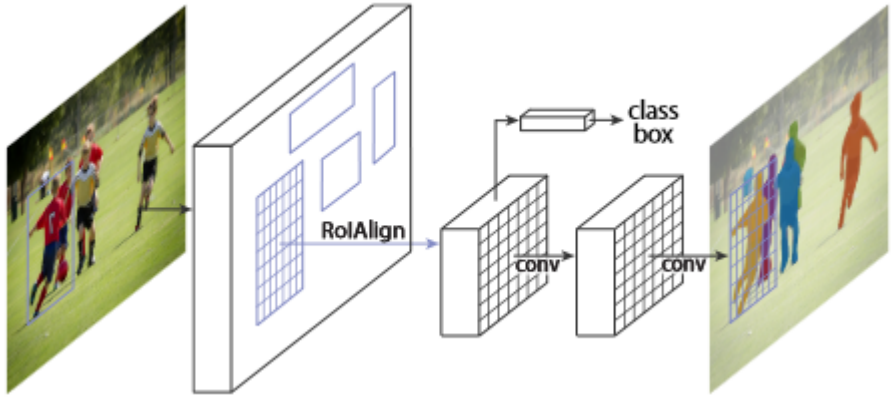


https://arxiv.org/pdf/1312.4659.pdf

Pose estimation – 2D

Mask R-CNN

- Can do pose estimation in addition to instance segmentation
- Looking for only a single pixel as foreground (each joint/keypoint)
- Predict *K* masks, *K* = number of keypoints





https://arxiv.org/pdf/1703.06870.pdf

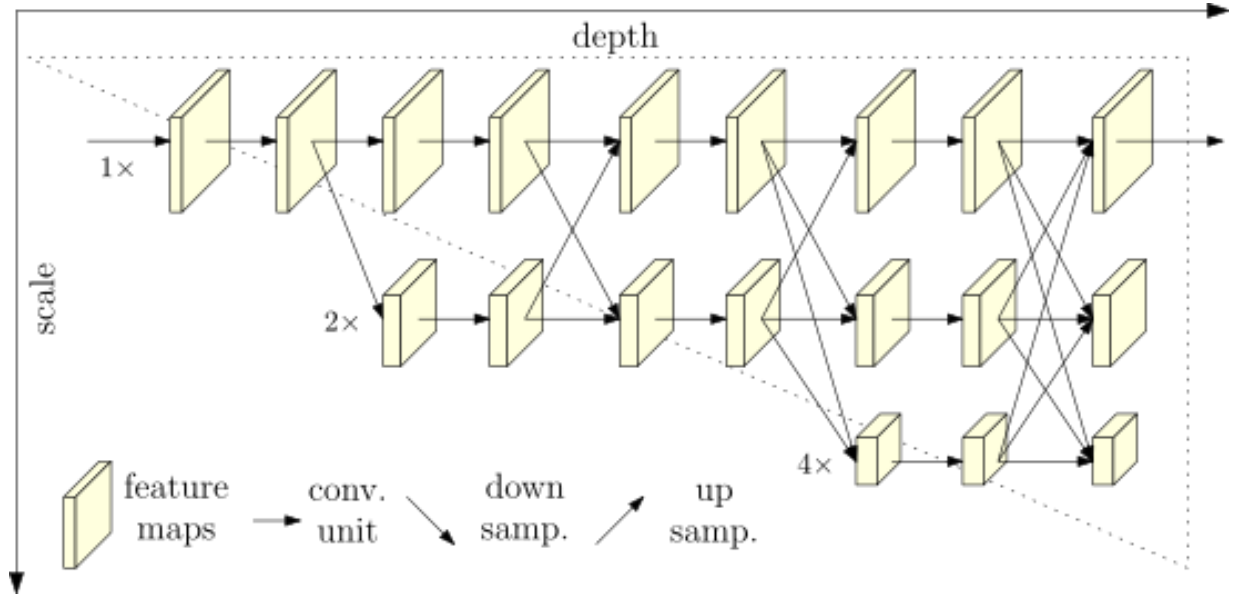
Pose estimation – 2D

Deep High-Resolution Representation Learning for Human Pose Estimation [HRNet]

- CVPR 2019

Idea:

- Instead of going high-low-high resolution, maintain a high resolution representation
- Parallel subnetworks



https://arxiv.org/pdf/1902.09212.pdf
https://github.com/leoxiaobin/deep-high-resolution-net.pytorch

Pose estimation – 2D

Evaluation example – HRnet

COCO dataset: 200,000 images, 250,000 labeled persons, 17 keypoints each

- Train on 57,000 images, 150,000 person instances
- Validation 5000 images, test 20,000 images
- Data augmentation: Rotation, scaling, flipping

Evaluation metrics (standardized by the COCO challenge):

- Object keypoint similarity [0,1]: $OKS = \frac{\sum_i \exp(-d_i^2/2s^2k_i^2)\delta(v_i>0)}{\sum_i \delta(v_i>0)}$
- d_i Euclidean distance between detected keypoint and corresponding ground truth
- v_i visibility flag of ground truth keypoint
- s object scale, k_i per-keypoint constant (sk_i standard deviation for Gaussian distribution)

https://arxiv.org/pdf/1902.09212.pdf
http://cocodataset.org/#keypoints-eval

Pose estimation – 2D

Evaluation metrics:

- $OKS = \frac{\sum_i \exp(-d_i^2/2s^2k_i^2)\delta(v_i>0)}{\sum_i \delta(v_i>0)}$

- OKS is similar to IoU for detection
- Used for Average Precision (AP) and Average Recall (AR)

```
Average Precision (AP):
  AP                  % AP at OKS=.50:.05:.95 (primary challenge metric)
  AP^OKS=.50          % AP at OKS=.50 (loose metric)
  AP^OKS=.75          % AP at OKS=.75 (strict metric)
AP Across Scales:
  AP^medium           % AP for medium objects: 32^2 < area < 96^2
  AP^large            % AP for large objects: area > 96^2
Average Recall (AR):
  AR                  % AR at OKS=.50:.05:.95
  AR^OKS=.50          % AR at OKS=.50
  AR^OKS=.75          % AR at OKS=.75
AR Across Scales:
  AR^medium           % AR for medium objects: 32^2 < area < 96^2
  AR^large            % AR for large objects: area > 96^2
```

http://cocodataset.org/#keypoints-eval



Pose estimation – 2D

Results:

- AP=75.5 on COCO test set



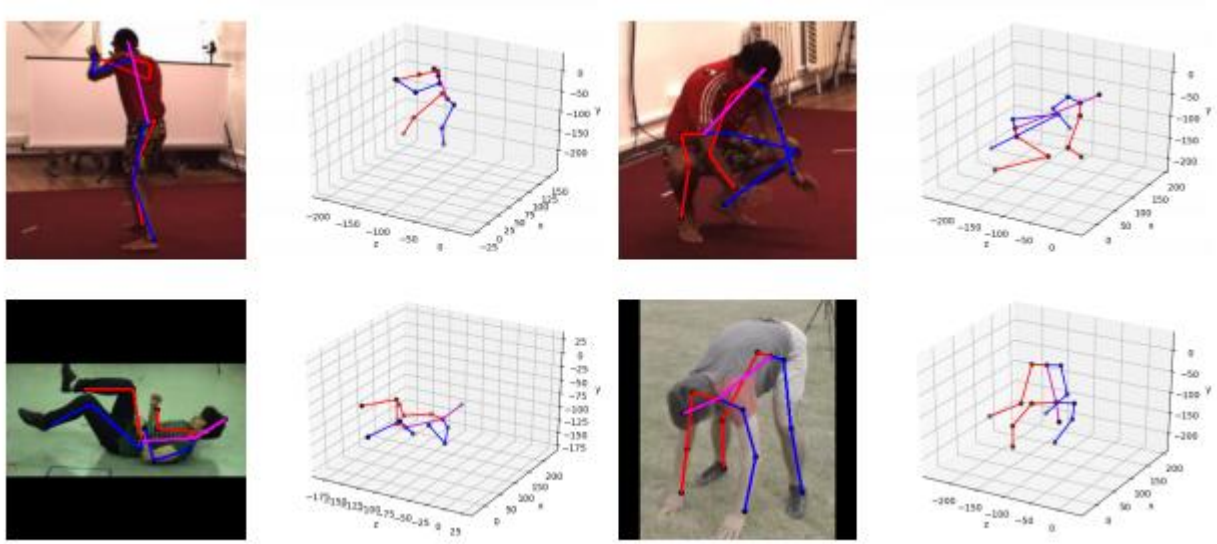
https://arxiv.org/pdf/1902.09212.pdf

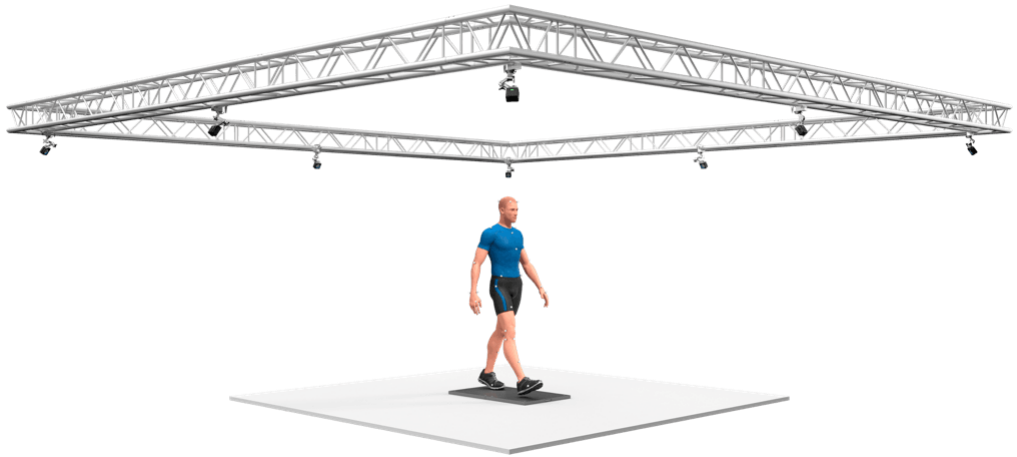
Pose estimation – 3D

Estimating joint positions in 3D space

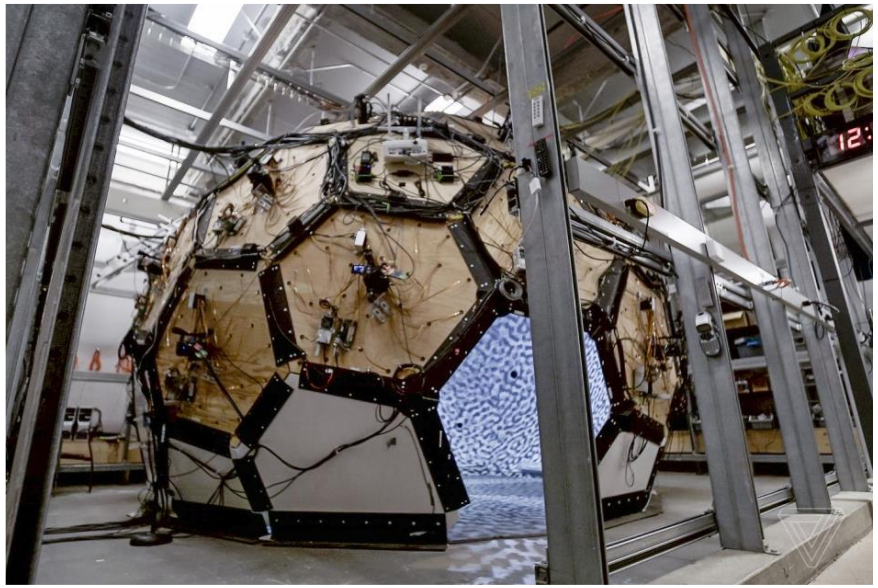
Ground truth:

- Optical motion capture system
- Reflective markers on joints
- Dome multiview system





Optitrack solution

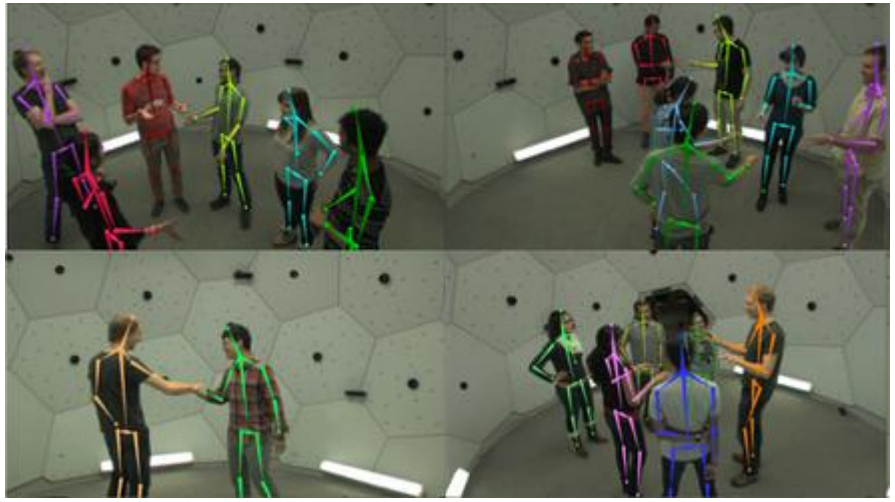


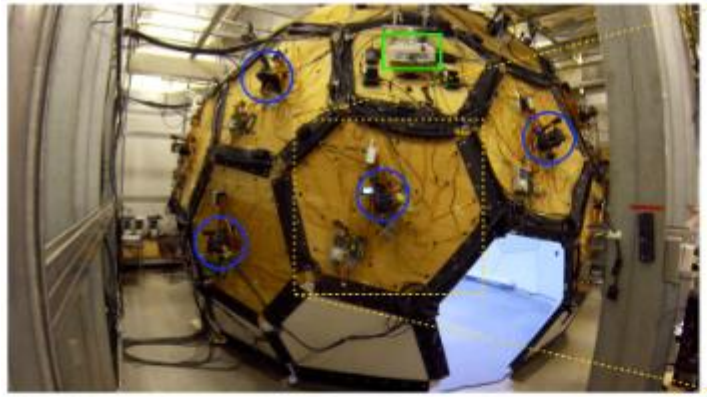
CMU Panoptic Studio

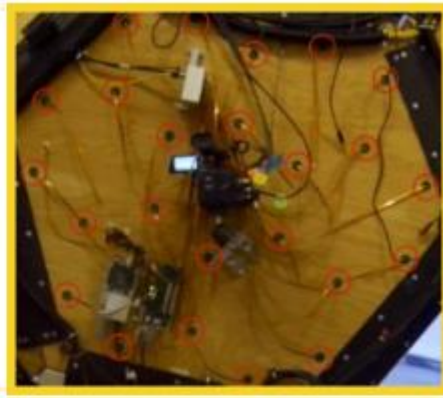
Pose estimation – 3D

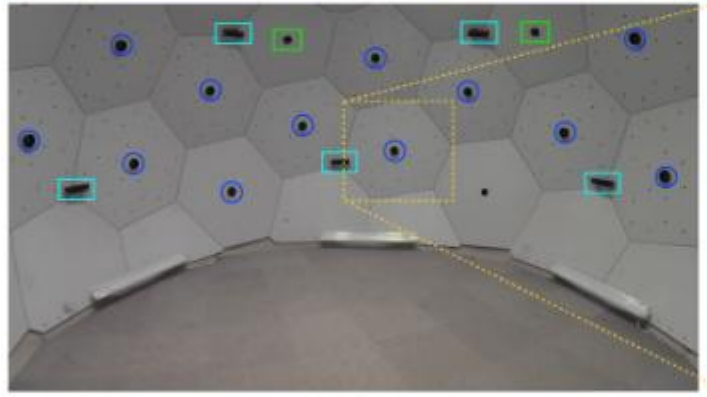
CMU Panoptic Studio Dataset:

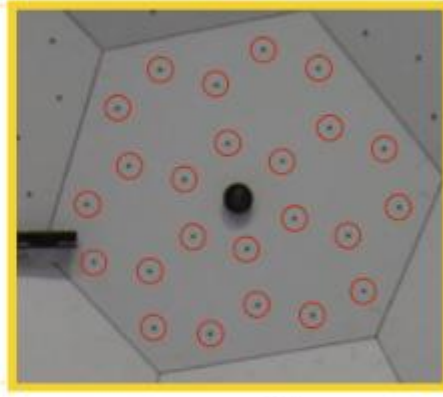
- Massively Multiview System
  - 480 VGA camera views
  - 30+ HD views
  - 10 RGB-D sensors
  - Hardware-based sync
  - Calibration

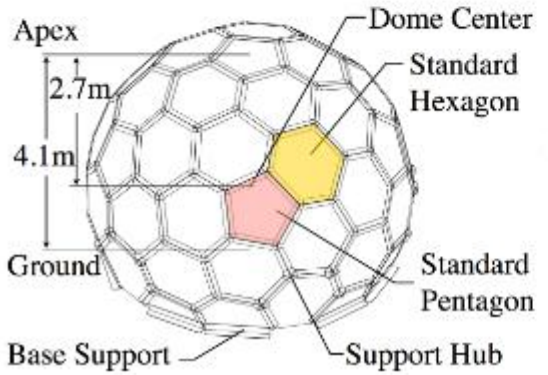


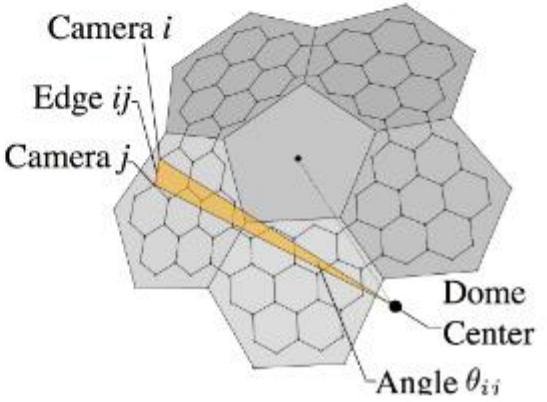












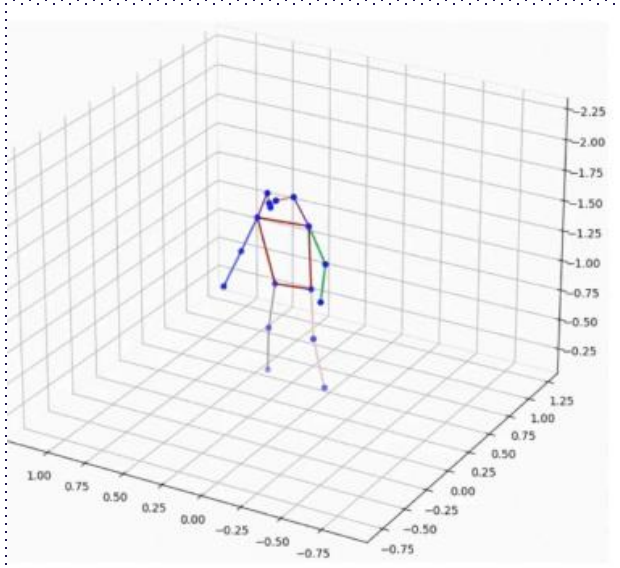
http://domedb.perception.cs.cmu.edu/

Pose estimation – 3D

SportsPose - A Dynamic 3D sports pose dataset (DTU and TrackMan)

- 7 hardware-synchronized cameras
- 90 Hz, 1920 x 1200 pixels
- 5 different sports activities



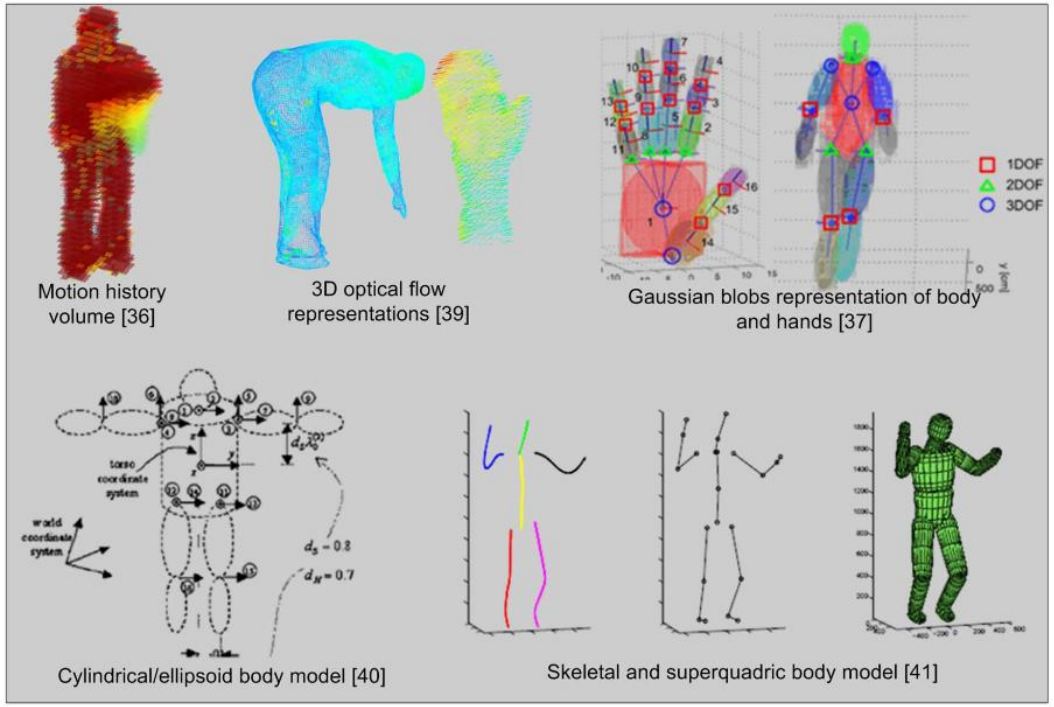


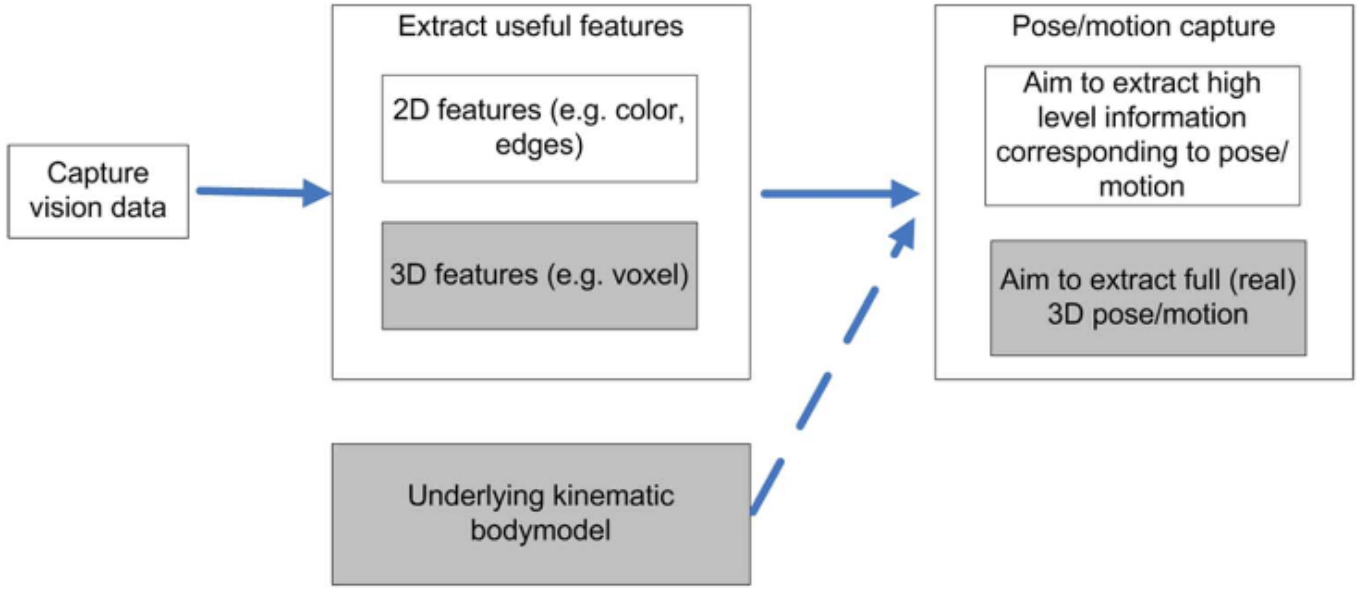
https://christianingwersen.github.io/SportsPose/

Pose estimation – 3D

Pre-deep learning – model-based approaches

Multi-view – input:
3D features (voxels) or multiple 2D views





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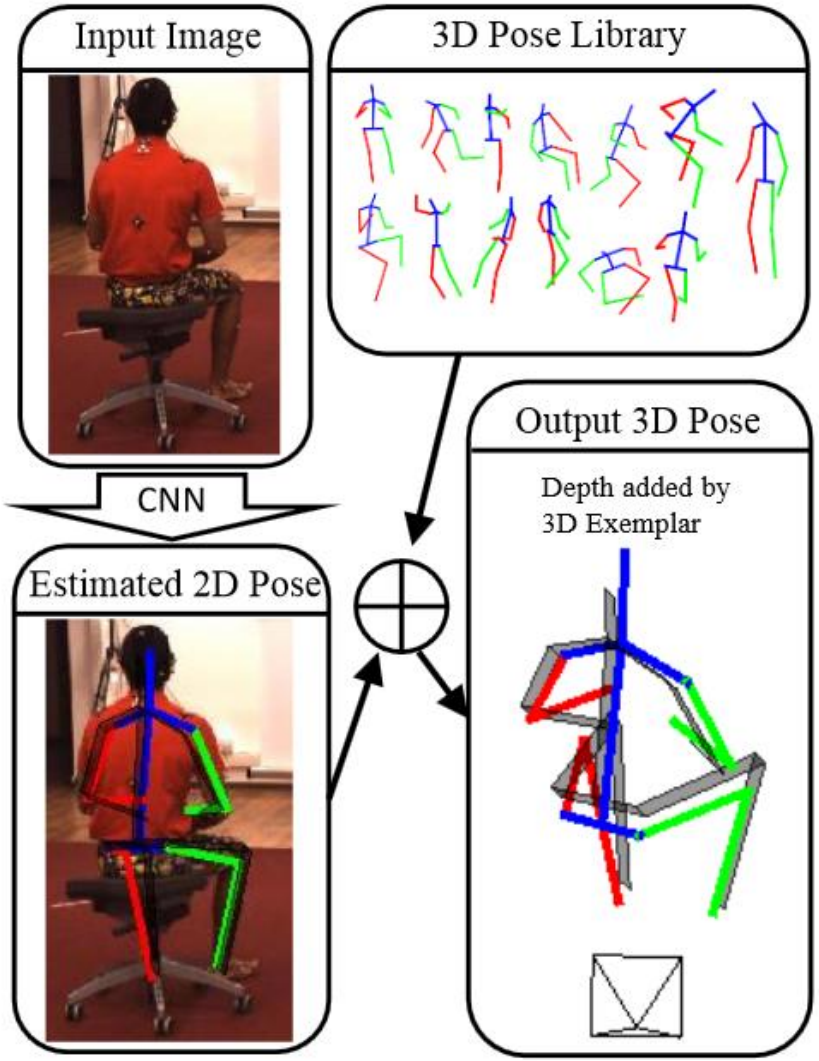
https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=6193117

Pose estimation – 3D

3D pose estimation from single image

- Deep learning approach

- Given a 3D pose library (MoCap or similar), generate a large number of 2D projections from virtual camera views
- Use this paired data (2D, 3D) as training set
- Given an input image, find 2D pose
- Then match to the library of 3D poses



https://arxiv.org/pdf/1612.06524.pdf

Pose estimation – implementations

OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields: https://github.com/CMU-Perceptual-Computing-Lab/openpose

MediaPipe Pose: https://developers.google.com/edge/mediapipe/solutions/vision/pose_landmarker

Yolo26-Pose: https://docs.ultralytics.com/tasks/pose

MMPose: https://github.com/open-mmlab/mmpose



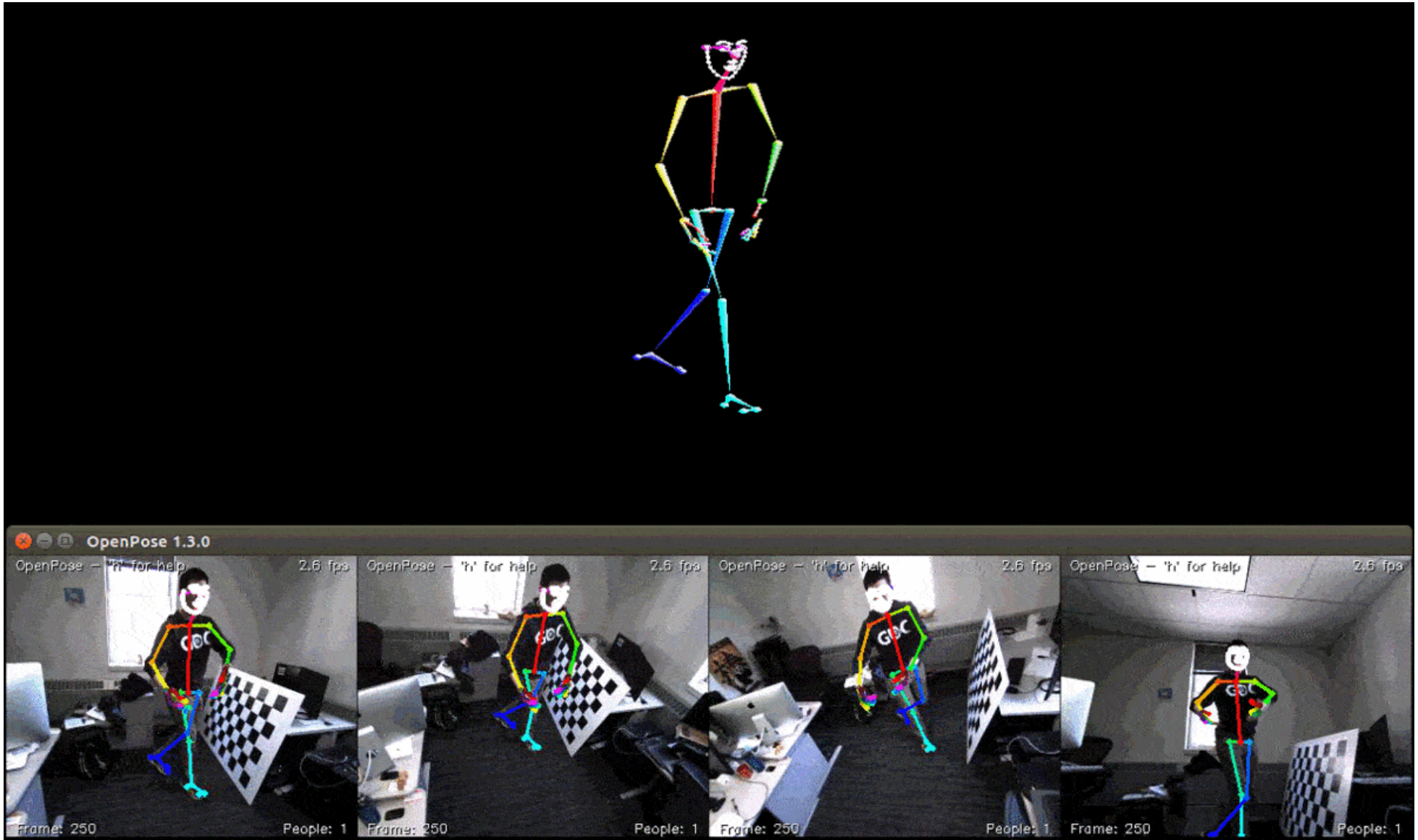
Fig. 1: **Top:** Multi-person pose estimation. Body parts belonging to the same person are linked, including foot keypoints (big toes, small toes, and heels). **Bottom left:** Part Affinity Fields (PAFs) corresponding to the limb connecting right elbow and wrist. The color encodes orientation. **Bottom right:** A 2D vector in each pixel of every PAF encodes the position and orientation of the limbs.

https://arxiv.org/pdf/1812.08008.pdf

OpenPose



OpenPose



Pose estimation approaches

In images with more than one person there are two main approaches:

Bottom up:

1. Detect first all keypoints across the image
2. Associate detected keypoints to person instances

Top-down:

1. Person detection
2. Single-person pose estimation in cropped regions

Summary

Human pose estimation

- Classic approaches: relations of body parts
- CNN based approaches: regression of joint positions
- 2D vs 3D
- 3D from 2D!



Next steps

Today:

- Three short videos on point cloud processing available on Moodle
- Exercises

Next weeks:

- Work on mini-projects in your own time
- Early November: Scheduled TA hours for mini-project (Vasiliki (Vikki) Ismiroglou will visit your group rooms).
  - If you get stuck before that, please email Vikki with your questions: vasilikii@create.aau.dk
- November 16: Hand-in mini-projects
- November 16-23: Review two mini-project solutions from other students
- November TBD: Depth sensing lecture, given by Stefan Hein Bengtson
- ARP only: November TDB: Camera-robot calibration, given by Stefan Hein Bengtson

